Sri Lanka Institute of Information Technology

IT2130 – Operating Systems and System Administration
Year 2, Semester 2- 2026

Practical Answer Submission

Practical Sheet No: 02

Student ID	IT24104150
Student Name	Sathsaranee K.L.U
Campus/ Center name	Malabe
Specialization	CSNE
Batch No	Y2S2.WD.CSNE.01

Activity 1

```
  GNU nano 7.2                          activity1.c
#include <stdio.h>

int main() {
    int mark1, mark2;
    float average;

    printf("Input the 1st mark: ");
    scanf("%d", &mark1);

    printf("Intput the 2nd mark: ");
    scanf("%d", &mark2);

    average = (mark1+mark2)/2.0;
    printf("The average is: %f\n", average);

    return 0;
}
```

Output:

```
$ nano activity1.c
$ gcc -oactivity1 activity1.c
$ ./activity1
Input the 1st mark: 20
Intput the 2nd mark: 30
The average is: 25.000000
```

Explain:

This program takes two integer marks from the user, calculates their average using floating-point division, and prints the result. Using 2.0 ensures the average includes decimal values.

Activity 2

```
  GNU nano 7.2                          activity2.c
#include <stdio.h>

int main() {
    int mark1, mark2;
    float average;

    printf("Input the 1st mark: ");
    scanf("%d", &mark1);

    printf("Intput the 2nd mark: ");
    scanf("%d", &mark2);

    average = (mark1+mark2)/2.0;
    printf("The average is: %f\n", average);

    if(average > 45){
        printf("Pass\n");
    }
    else {
       printf("Fail\n");
    }

    return 0;
}
```

Output:

```
$ gcc -oactivity2 activity2.c
$ ./activity2
Input the 1st mark: 45
Intput the 2nd mark: 60
The average is: 52.500000
Pass
```

Activity 3

```
  GNU nano 7.2                                                                  activity3.c
#include <stdio.h>

int main() {
    char gender;
    int age;

    printf("Enter gender (M/F): ");
    scanf(" %c", &gender);

    printf("Enter age: ");
    scanf("%d", &age);

    if (age >= 65) {
        if (gender == 'M' || gender == 'm') {
            printf("Senior Male citizen\n");
        } else if (gender == 'F' || gender == 'f') {
            printf("Senior Female citizen\n");
        } else {
            printf("Invalid gender\n");
        }
    } else {
        printf("Not a senior citizen\n");
    }

    return 0;
}
```

Output:

```
Enter gender (M/F): m
Enter age: 70
Senior Male citizen
```

Activity 4

```
  GNU nano 7.2                                                                  activity4.c
#include <stdio.h>

int main() {
    int mark;

    printf("Enter the mark: ");
    scanf("%d", &mark);

    if (mark >= 80 && mark <= 100) {
        printf("Grade: A\n");
    } else if (mark >= 70 && mark <= 79) {
        printf("Grade: B\n");
    } else if (mark >= 45 && mark <= 69) {
        printf("Grade: C\n");
    } else if (mark < 45 && mark >= 0) {
        printf("Grade: F\n");
    } else {
        printf("Invalid mark entered\n");
    }

    return 0;
}
```

Output:

```
Enter the mark: 70
Grade: B
```

Activity 5

```
  GNU nano 7.2                                  activity5.c
#include <stdio.h>

int main() {
    int choice;

    printf("Menu:\n");
    printf("1. print \"Hello\"\n");
    printf("2. print \"Welcome\"\n");
    printf("3. print \"Goodbye\"\n");

    printf("Enter your choice (1 to 3): ");
    scanf("%d", &choice);

    switch (choice) {
        case 1:
            printf("Hello\n");
            break;
        case 2:
            printf("Welcome\n");
            break;
        case 3:
            printf("Goodbye\n");
            break;
        default:
            printf("Invalid choice\n");
    }
    return 0;
}
```

Output:

```
Menu:
1. print "Hello"
2. print "Welcome"
3. print "Goodbye"
Enter your choice (1 to 3): 2
Welcome
```

Activity 6

```
  GNU nano 7.2                                                    activity6.c
#include <stdio.h>

int main() {
    int mark;
    char grade;

    printf("Enter the mark: ");
    scanf("%d", &mark);

    if(mark >= 80 && mark <= 100){
       grade = 'A';
    }
    else if(mark >= 70 && mark <= 79){
       grade = 'B';
    }
    else if(mark >= 45 && mark <= 69){
       grade = 'C';
    }
    else if(mark >= 0 && mark < 45){
      grade = 'F';
    }
    else {
      grade = 'X';
    }

     switch (grade) {
         case 'A':
            printf("Grade: A\n");
            break;
        case 'B':
            printf("Grade: B\n");
            break;
        case 'C':
            printf("Grade: C\n");
            break;
       case 'F':
            printf("Grade: F\n");
            break;
       default:
            printf("Invalid mark\n");
     }
     return 0;
}
```

Output:

```
Enter the mark: 78
Grade: B
```